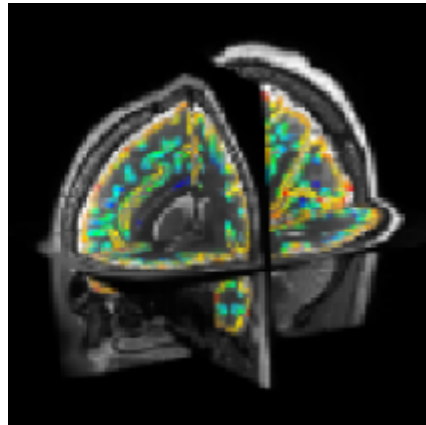
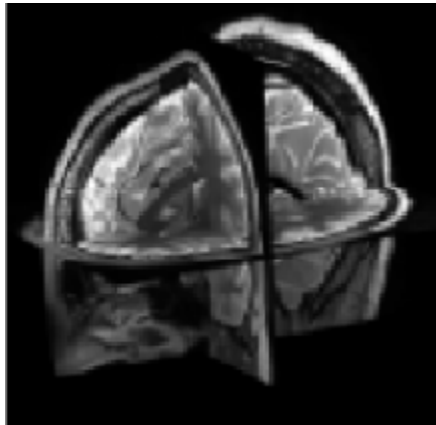


Use Case 4: Mapping cortical structures in brain MRI



Problem category: Semantic Segmentation

Recommended metrics:

- Overlap-based Metric: [Center line Dice Similarity Coefficient](#)
- Boundary-based Metric: [Normalized Surface Distance \(NSD\)](#)

Fingerprint answers:

Fingerprint	Answer
Small size of structures relative to pixel size	No
Particular importance of structure center (e.g. in cells, vessels)	Yes
Target structures feature tubular shape	Yes
Penalization of errors: Unequal severity of class confusions	Equal handling
Possibility of overlapping or touching target structures (e.g. medical instruments or cells)	No
Particular importance of structure boundaries	Yes
Penalization of errors: Compensation for annotation imprecisions requested	Yes